
Highly Available 3-Tier Web Application on AWS

Project Overview

This project demonstrates the deployment of a **highly available, scalable, and fault-tolerant web application** using AWS cloud services. The architecture is designed to handle traffic spikes and ensure high availability across multiple Availability Zones.

Architecture Description

The application follows a **3-tier architecture**:

Internet



Application Load Balancer (ALB)



Target Group



EC2 Instances (Auto Scaling Group)



EBS (Application Data)



S3 (Static Assets)

Additional Components:

- Bastion Host (for secure SSH access)
 - AWS Backup (for EBS snapshots)
-

AWS Services Used

- Amazon EC2 – Virtual servers for application hosting
- Application Load Balancer – Distributes incoming traffic
- Amazon S3 – Stores static assets

- Auto Scaling – Automatically scales instances
- Amazon EBS – Persistent storage
- AWS Backup – Snapshot automation
- AWS Certificate Manager – HTTPS security

1. VPC & Networking Setup

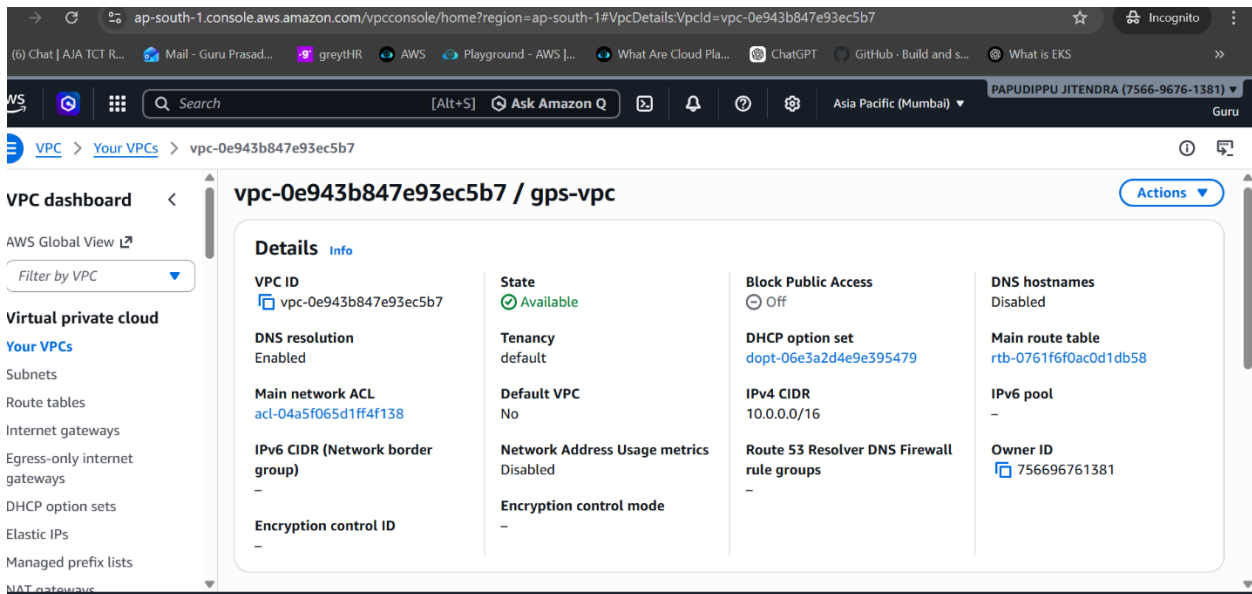
Objective:

Create a secure and isolated network.

Steps:

1. Created a **VPC** with CIDR:

10.0.0.0/16

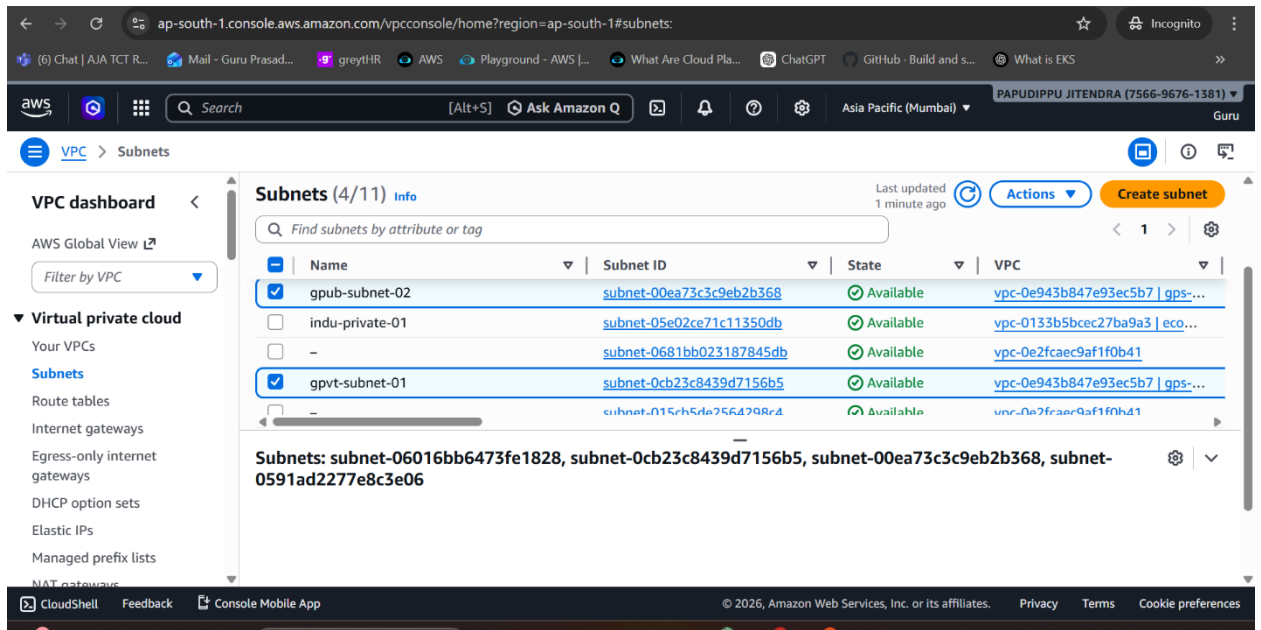


The screenshot shows the AWS VPC console for a VPC with ID vpc-0e943b847e93ec5b7, named 'gps-vpc'. The console displays various configuration details in a grid layout.

vpc-0e943b847e93ec5b7 / gps-vpc			
VPC ID vpc-0e943b847e93ec5b7	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-06e3a2d4e9e395479	Main route table rtb-0761f6f0ac0d1db58
Main network ACL acl-04a5f065d1ff4f138	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 756696761381
Encryption control ID -	Encryption control mode -		

2. Created **Subnets in 2 Availability Zones:**

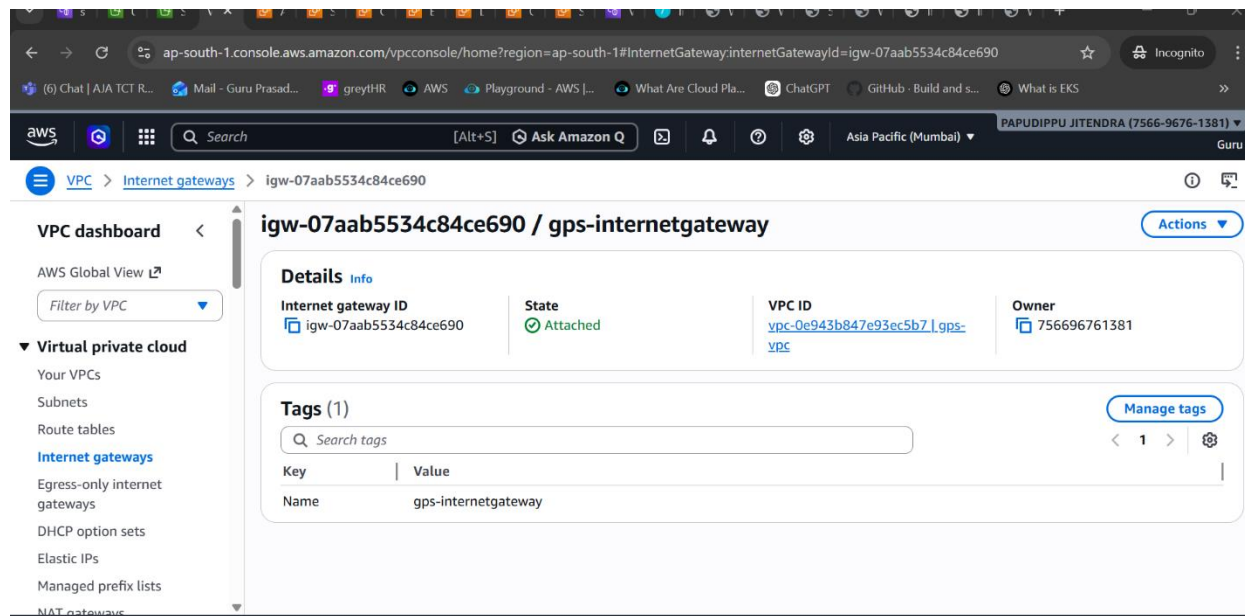
- Public Subnet (for ALB & Bastion)
- Private Subnet (for EC2)



3.

4. Attached:

- Internet Gateway (for public access)



5. Configured Route Tables:

- Public → Internet Gateway

6.

6.

- Private → internal routing

7.

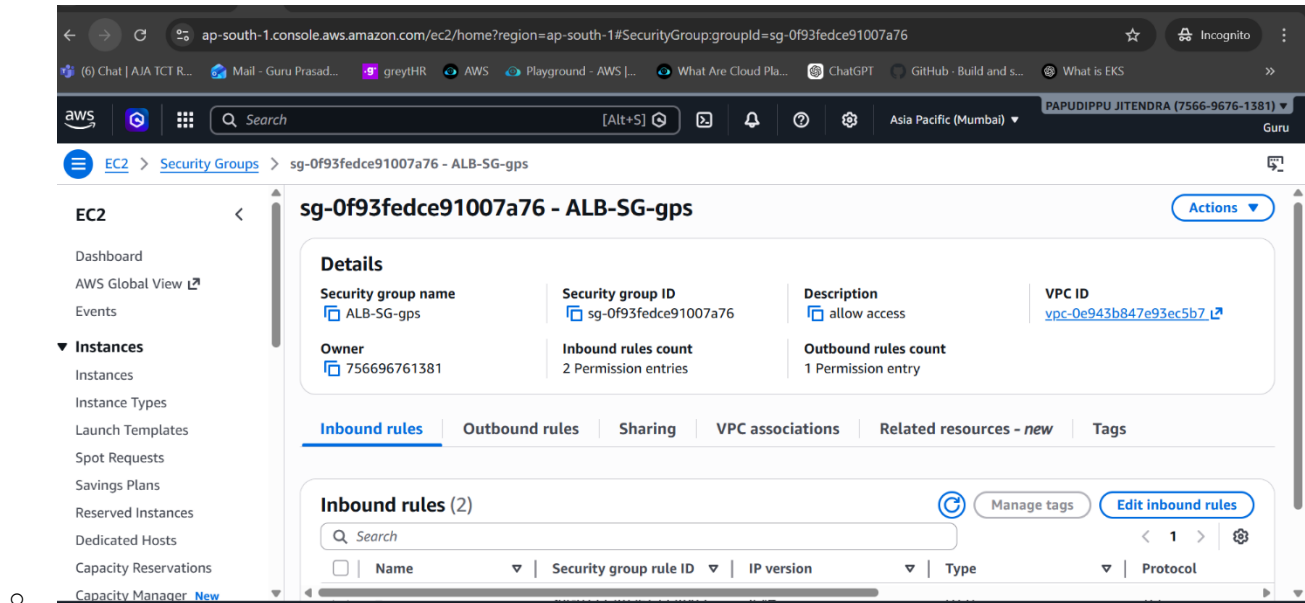
7.

2. Security Groups Configuration

ALB Security Group:

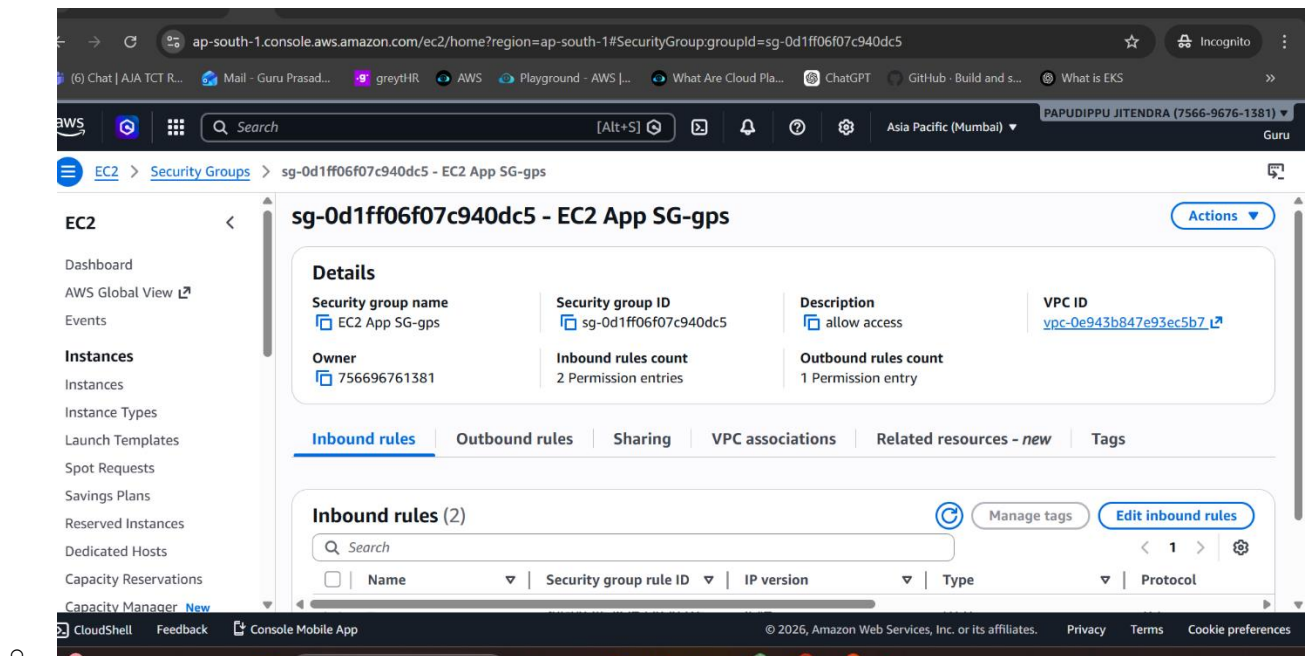
- Allow:

- HTTP (80) from anywhere
- HTTPS (443) from anywhere



EC2 Security Group:

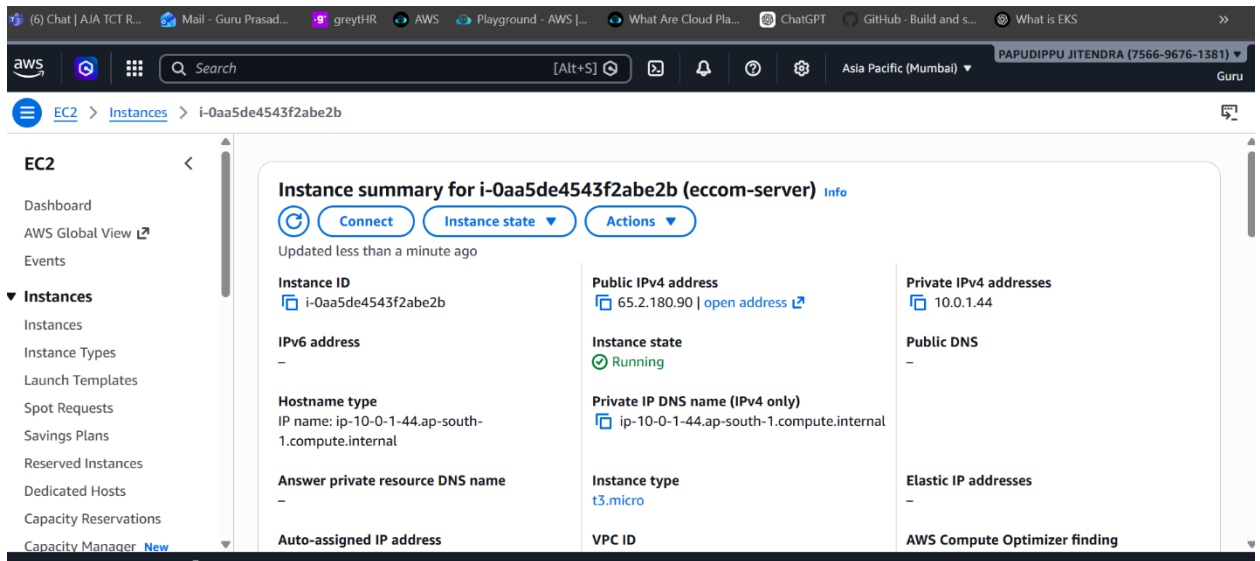
- Allow:
 - HTTP (80) from ALB Security Group
 - SSH (22) from Bastion Host



3. EC2 & Golden AMI

Steps:

1. Launched EC2 instance



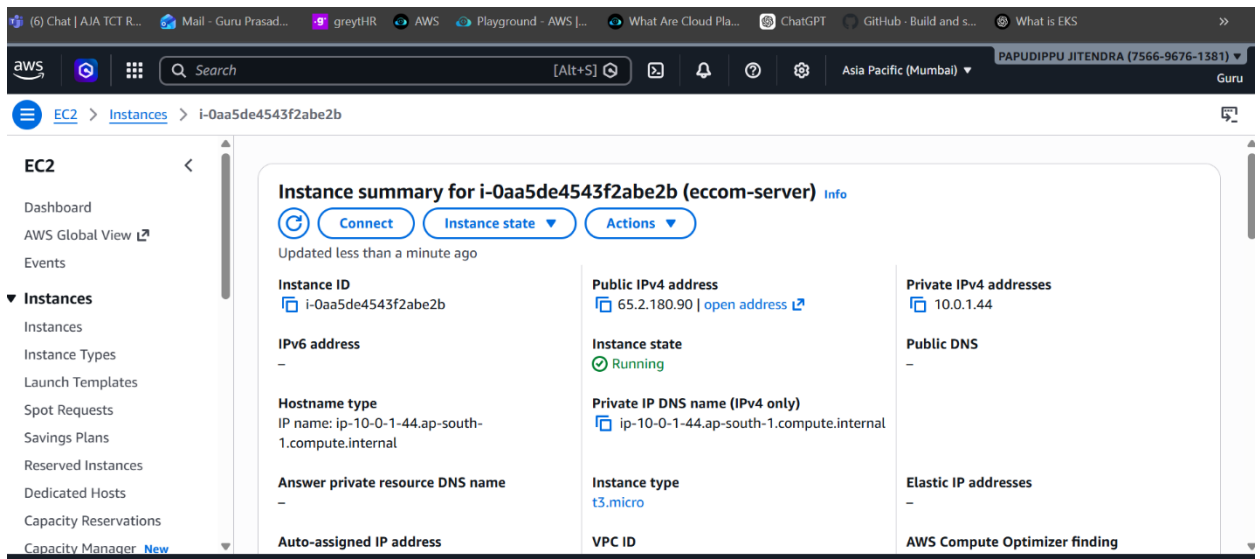
2.
3.

4. Installed web server:

`sudo apt install nginx -y`

Created **Golden AMI**

- Used for launching identical instances

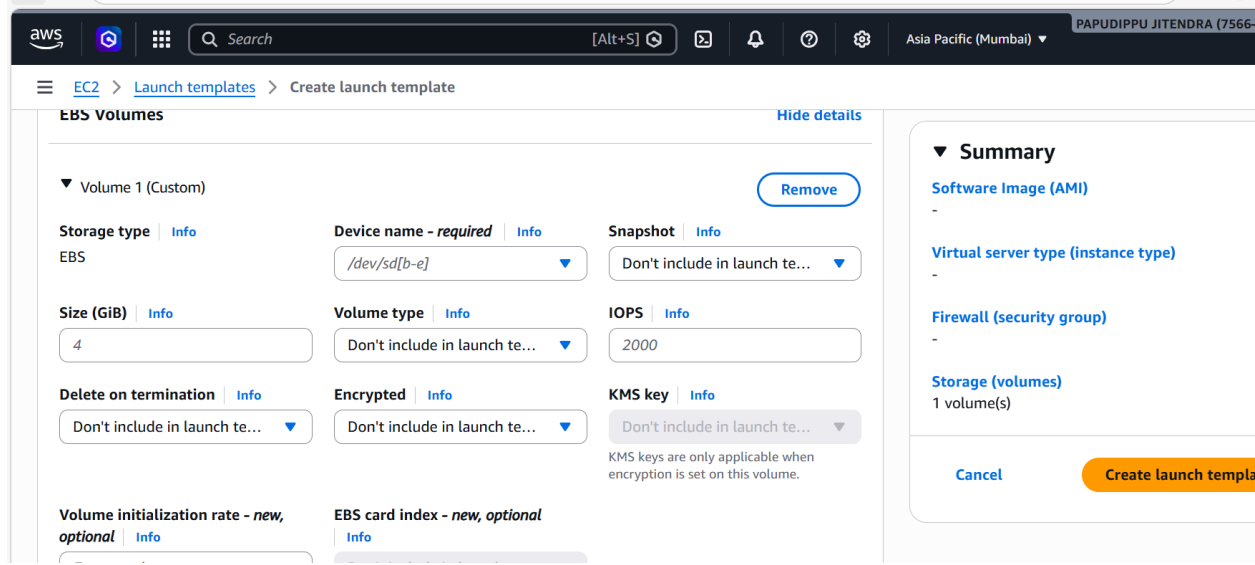


5.

○

4. EBS Storage

- Attached **gp3 volume (20GB)** to EC2
- Enabled:
 - Encryption using KMS



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EC2 > Launch templates > Create launch template

EBS Volumes [Hide details](#)

Volume 1 (Custom) [Remove](#)

Storage type [Info](#)
EBS

Device name - required [Info](#)
/dev/sd[b-e]

Snapshot [Info](#)
Don't include in launch te...

Size (GiB) [Info](#)
4

Volume type [Info](#)
Don't include in launch te...

IOPS [Info](#)
2000

Delete on termination [Info](#)
Don't include in launch te...

Encrypted [Info](#)
Don't include in launch te...

KMS key [Info](#)
Don't include in launch te...
KMS keys are only applicable when encryption is set on this volume.

Volume initialization rate - new, optional [Info](#)

EBS card index - new, optional [Info](#)

Summary

Software Image (AMI)
-

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
1 volume(s)

[Cancel](#) [Create launch template](#)

5. Target Group Configuration

Key Settings:

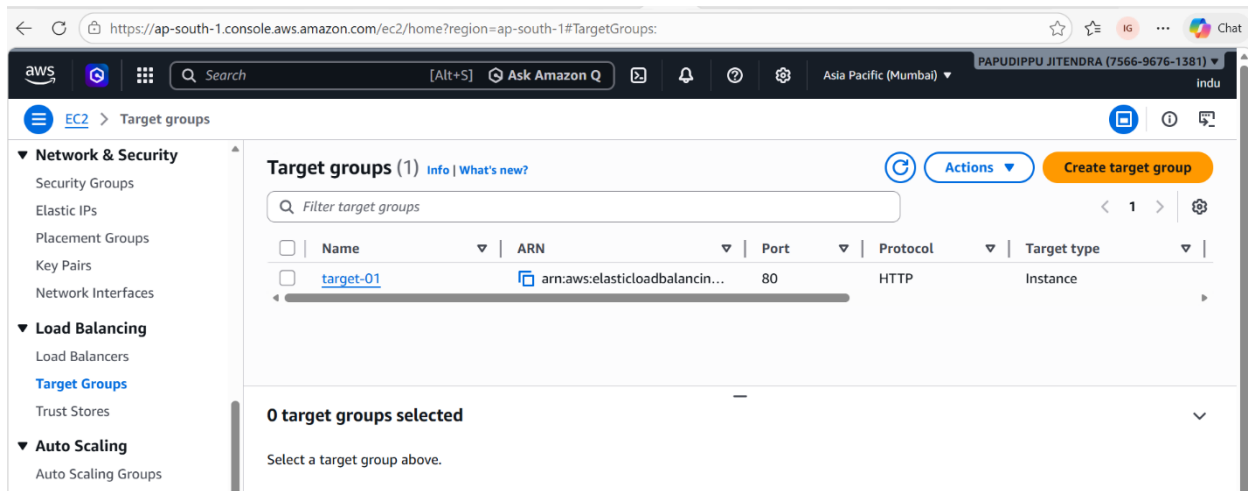
- Target Type: Instance
- Protocol: HTTP
- Port: 80
- Health Check Path:

/

Issue Faced:

- Initially:

0 Targets Registered



- Solution:
 - Manually registered EC2 instance
 - Verified health status

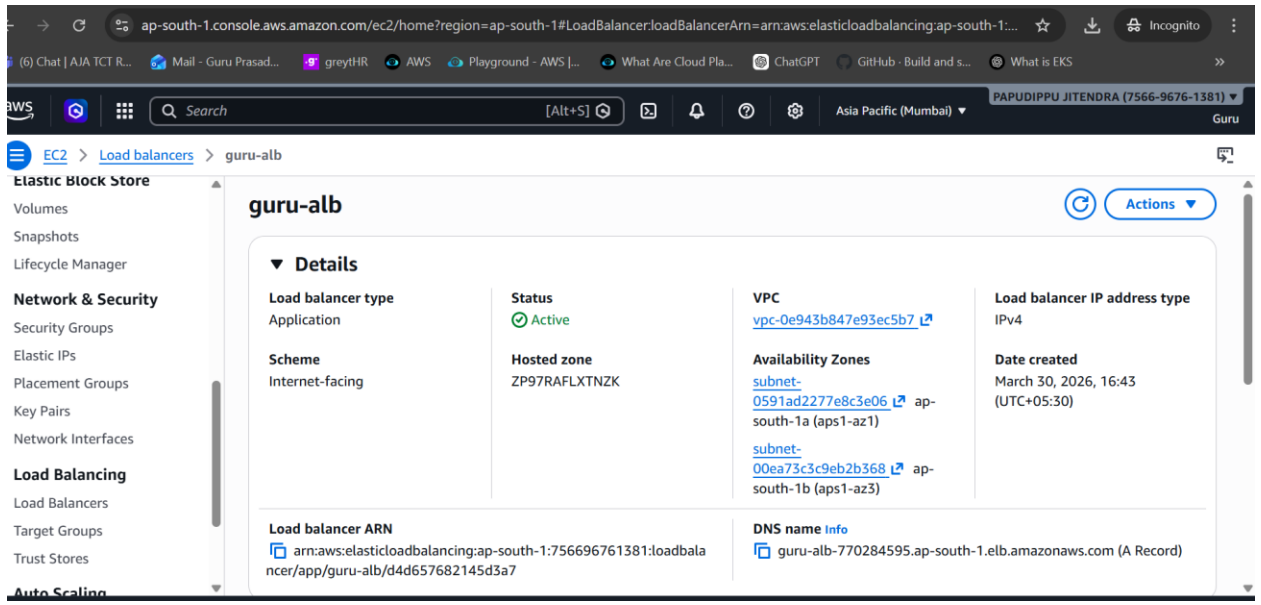
6. Application Load Balancer

Configuration:

- Type: Internet-facing
- Subnets: Public (2 AZs)
- Listener:
 - HTTP (80) → Target Group

Features Enabled:

- Cross-Zone Load Balancing
- Sticky Sessions (optional)



7. Auto Scaling Group

Configuration:

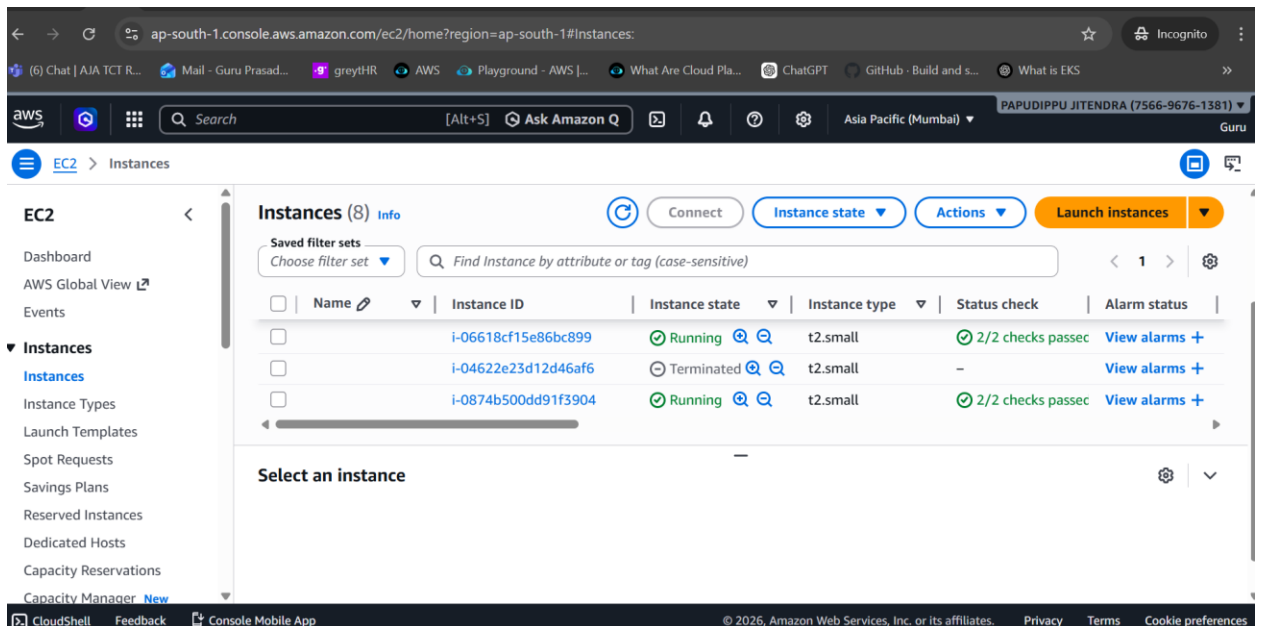
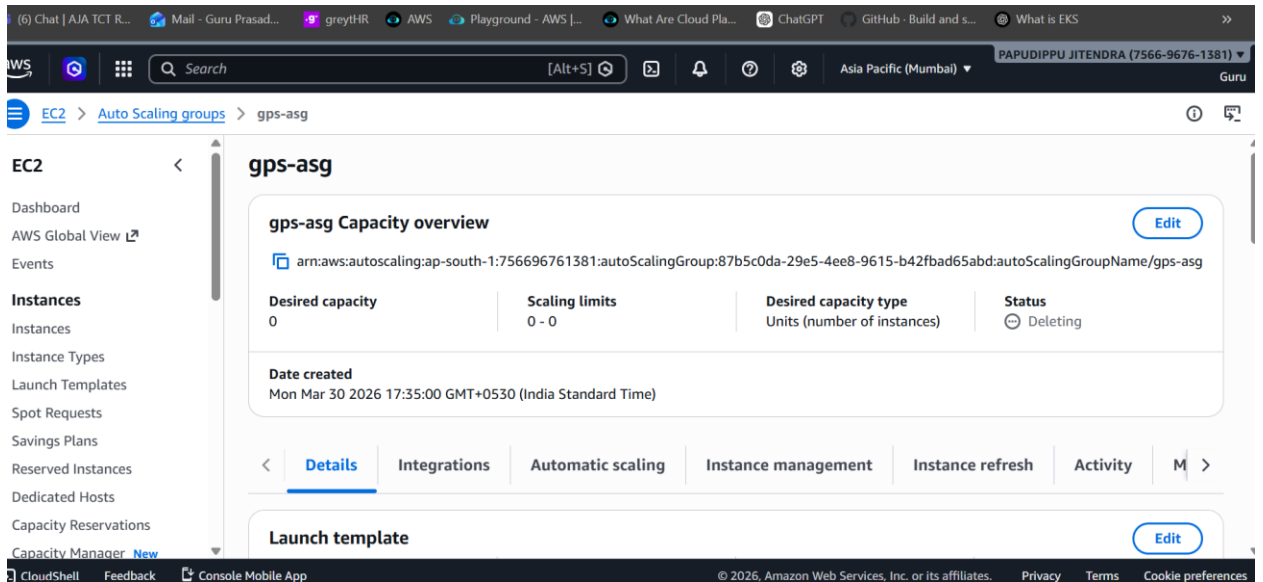
- Launch Template used (Golden AMI)
- Min: 2
- Desired: 2
- Max: 6

Scaling Policy:

- Scale out: CPU > 60%
- Scale in: CPU < 30%
- Cooldown: 300 seconds

Important:

- Attached Target Group to ASG
- Ensures automatic registration of instances

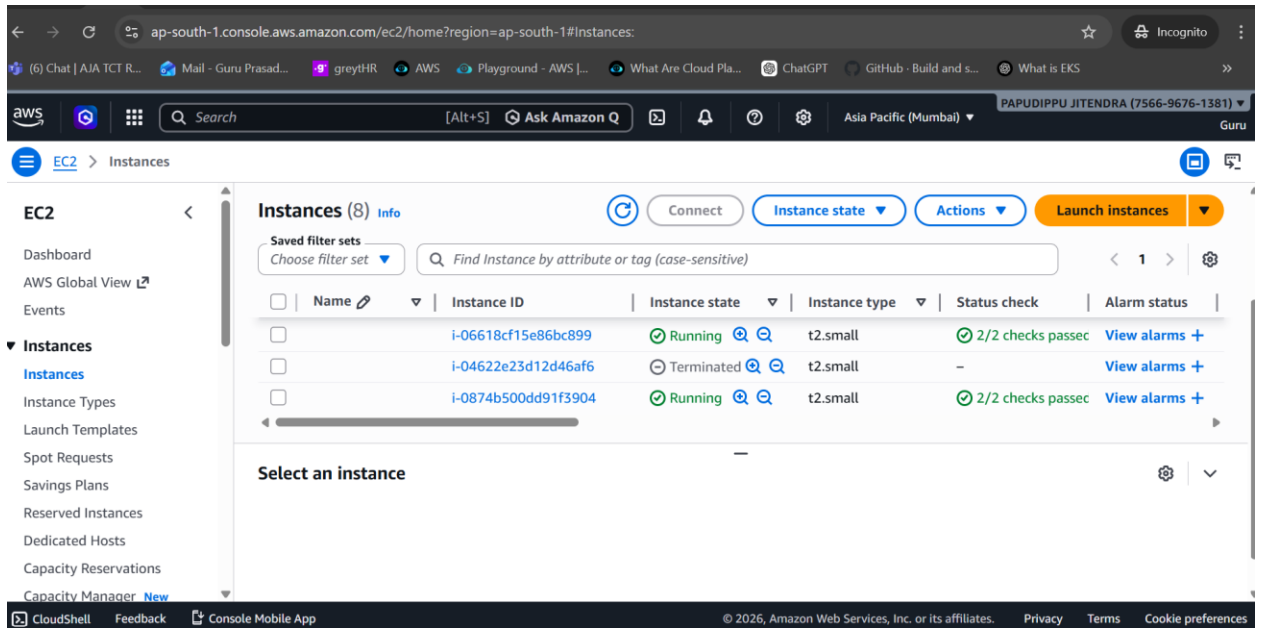


8. S3 Bucket (Static Assets)

Configuration:

- Enabled:

- Versioning
- Encryption (SSE-S3)



Usage:

- Stored:
 - Images
 - CSS
 - JS
 -

← → ↻ ap-south-1.console.aws.amazon.com/s3/buckets/eccom-task?region=ap-south-1&tab=objects ☆ Incognito

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Amazon S3 Buckets > eccom-task

Amazon S3

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

▼ Storage management and insights

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (1)

⌂ Copy S3 URI

📄 Copy URL

⬇ Download

🔗 Open

🗑 Delete

⋮ Actions

Create folder

⬆ Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 > ⚙

☐	Name	Type	Last modified	Size	Storage class
☐	 AWS-serviceslist.jpg	jpg	March 30, 2026, 17:48:48 (UTC+05:30)	101.1 KB	Standard

CloudShell Feedback Console Mobile App

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